Scalar Curvature Rigidity 2

Thm 1 (Llarull '98)

(M^n, g) closed connected spin, $n \geq 2$

a) $\mathrm{scal}_g \geq n(n-1) = \mathrm{scal}_{g_{S^n}}$

b) $f : (M, g) \longrightarrow (S^n, g_0)$ smooth, $\deg(f) \neq 0$ s.t.

- f 1-Lipschitz, $n = 2$
- f area-nonincreasing, $n \geq 3$

$\Rightarrow f$ Riem. isometry

- **Goette-Semmelmann** '02: replace S^n by mfd. with positive Riem. curvature operator and non-vanishing Euler charakteristic
- **Lott** '21: Goette-Semmelmann for compact mfds. with boundary
- **Brendle** '24: Convex polytopes
- **Bär-Brendle-Hanke-Wang** '24: S^n with antipodal points removed

n even n odd

Thm 2 (Cecchini-Hanke-Schick '22, C-H-S-S '25)

M^n smooth closed connected spin, $n \geq 2$

a) $g \in W^{1,p}(T^*M \otimes T^*M)$, $p > n$, admissible Riem. metric s.t.

$$\text{scal}_g \geq_{\text{distr.}} n(n-1)$$

b) $f : (M, g) \longrightarrow (S^1, g_0)$ Lipschitz, $\deg(f) \neq 0$ s.t.

- f 1-Lipschitz, $n = 2$
- f area-nonincreasing a.e., $n \geq 3$

$\Rightarrow f : (M, d_g) \longrightarrow (S^1, d_0)$ metric isometry

Remark Lee-Tam '22: Thm 2 in the case f is 1-Lipschitz $\forall n \geq 2$

Def M closed mfd.

- $0 < \vartheta \leq 1$, $g_r \in C^\infty((0, \vartheta), C^\infty(T^*M \otimes T^*M))$ family of Riem. metrics
- $g_0 \in C^2(T^*M \otimes T^*M)$

s.t. $g_r \xrightarrow{r \to 0} g_0$ in C^2 and

$$|r\partial_r g_r| \longrightarrow 0 \quad , \quad |r^2 \partial_r^2 g_r| \longrightarrow 0 \quad \text{as} \quad r \to 0$$

Then $Cg_r := dr^2 + r^2 g_r$ on $(0,\vartheta) \times M$ is called generalized cone metric.

(W, G) is a smooth Riem. mfd. with cone-like singularities, if

- $\exists\, K \subset W$ compact
- $\exists\, 0 < \vartheta \leq 1$ and Riem. isometry

$$\upsilon : (W \setminus K, G|_{W\setminus K}) \longrightarrow ((0,\vartheta) \times \partial K, Cg_r)$$

(Cg_r: generalized cone metric)



Thm 3 (C-H-S-S '25)

(W^{n+1}, G) Riem. spin mfd. with cone-like sing. $n \geq 3$ odd

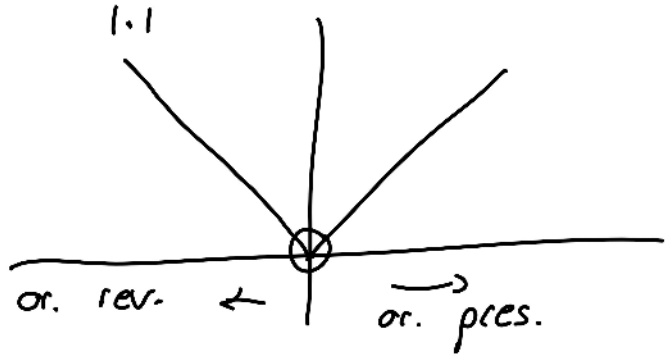
- $\mathrm{scal}_G \geq (n+1)n$
- $f : W \longrightarrow S^{n+1}$ Lipschitz, area-nonincreasing a.e. $\deg(f) \neq 0$

$\Rightarrow$ f Riem. isometry onto a punctured sphere

Proof of Thm 2 part 2

local isometry $\rightsquigarrow$ metric isometry for n even

to show: f either or. pres. or or. rev. a.e.



$\rightarrow$ l.l local isometry a.e. but

$$2 = d(-1,1) \neq d(|-1|, |1|) = 0$$

$0 \neq u \in \ker(D_E^+)$ $\qquad \Sigma M = \Sigma^+ M \oplus \Sigma^- M$

From SL-formula

- $(e_i \cdot e_j \otimes f_* e_i \cdot f_* e_j) \cdot u = u$ (1)
- $\nabla^{S \otimes E} u = 0$ a.e. $\rightarrow$ $|u| = \text{const.}$ a.e. (2)

Let $x \in M$ s.t. $d_x f$ exists

Let $\omega_{\mathbb{C},M}$, $\omega_{\mathbb{C},S^n}$ be the volume forms

$i^k e_1 \cdot \ldots \cdot e_n$

$\Rightarrow (\omega_{\mathbb{C},M} \otimes \omega_{\mathbb{C},S^n}) \cdot u = u \qquad (3)$

f local isometry a.e. $\Rightarrow \omega_{\mathbb{C},S^n} = \pm i^k f_* e_1 \cdot \ldots \cdot f_* e_n$

$+$ if f or. pres.

$-$ $-$ " or. rev.

$$\Rightarrow \pm u \overset{(3)}{=} i^{2k} (e_1 \cdot \ldots \cdot e_n \otimes f_* e_1 \cdot \ldots \cdot f_* e_n) \cdot u$$

$$\overset{(1)}{=} (-1)^k u$$

$\Rightarrow$ f orient $\begin{cases} \text{pres.} & \text{in } \times \text{ if } k \text{ even} \\ \text{rev.} & - " - \quad k \text{ odd} \end{cases}$

$\overset{(2)}{\Rightarrow}$ holds a.e.

$\Rightarrow$ f metric isometry

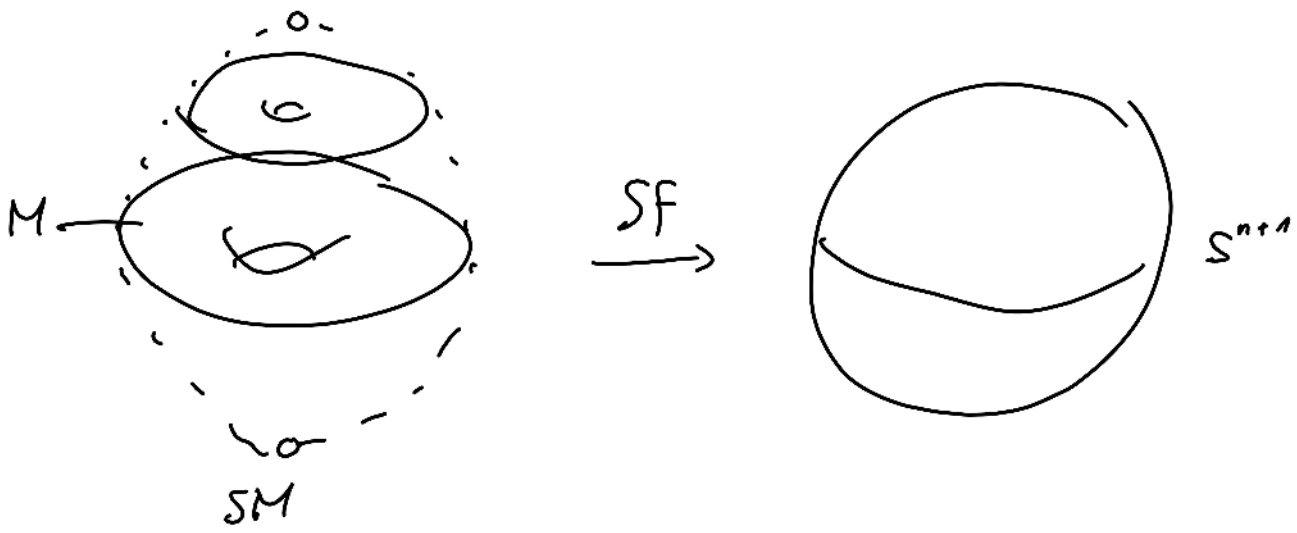
(Reshetnyak

n odd

Consider the spherical susp.

$SM := M \times (0, \pi) \qquad Sg := dr^2 + \sin^2(r) g$

$Sf : (SM, Sg) \longrightarrow (SS^n, Sg_n) = (S^{n+1} \setminus [N,S], g_{S^{n+1}})$



$$scal_g \geq (n+1)n = scal_{g_{S^{n+1}}}$$

<u>But</u> f area-nonincr. does not imply that

Sf — " —

<u>Solution</u>: first show f 1-Lipschitz using spectral flow

Then generalize even dim. proof to $\partial M \xrightarrow{Sf} S^{n+1}$

Caveat: Only have SL-formula for $p > n+1$ on ∂M

<u>Spectar flow</u>

"Def" Given a "continuous" path of self-adj. Fredholm op. D_t, $t \in [a,b]$, the spectral floco is

$sf(D_t)$ ":=" # { eigenvalues passing from − to + }

— # { − " — + to − }

Thm (Getzler '93)

M^{2k+1} closed connected spin, D Dirac op. on $\Sigma M \otimes \mathbb{C}^N$ for $N \in \mathbb{N}$

Let $U : M \to U(N)$ smooth and

$$D_t := (1-t)D + t\, U^{-1} D U \quad , \; t \in [0,1]$$

$$\rightsquigarrow \quad sf(D_t) = -\frac{1}{(-2\pi i)^{k+1}} \int_M \hat{A}(TM) \wedge Ch(U)$$

If $M = S^{2k+1}$ $\quad sf(D_t) = \deg(U)$

Def For $u \in C^\infty(\Sigma M)$ set, $X \in TM$

$$\nabla^t_X u = \nabla^{\Sigma M}_X u + t\, X \cdot u$$

∇^t-parallel spinors are called Killing spinors

Example: $n = 2k$, $n = 2k+1$

For $t = \pm \frac{1}{2}$, ΣS^n can be trivialized by $\nabla^{\pm\frac{1}{2}}$-parallel spinors i.e.

$$\exists \; \Sigma S^n \xrightarrow{V_\pm}_{\sim} S^n \times \mathbb{C}^{2^k} \qquad (*)$$

$f : M \to S^n$, consider $E = f^*(S^n \times \mathbb{C}^{2^k}) = M \times \mathbb{C}^{2^k}$ with trivial connection

$\rightsquigarrow$ D_E twisted Dirac op.

(*) induce trivializ. $U_\pm : \bar{E} \xrightarrow{\sim} M \times \mathbb{C}^{2k}$

Set $U = U_+ U_-^{-1} : M \longrightarrow U(\mathbb{C}^{2k})$

Li-Su-Wang

$\Rightarrow \quad sf(D_t) = \deg(f) \neq 0$

$\Rightarrow$ for some $t \in [0,1]$ $\exists\, u \in \ker(D_t)$

Abstract cone operators

Consider $SM = (0,\bar{r}) \times M$, $Sg = dr^2 + \sin^2(r)\, g$

There are bundle isometries

$$\Sigma^{\pm} SM \big|_{M \times \{x\}} \cong \Sigma M$$

Let $U = (0,1) \times M$

Using parallel transport of sections in $C^\infty(\Sigma M)$ along geodesic lines $(0,1) \times \{x\} \subset SM$, we get isometries



$$L^2(U, \Sigma^{\pm} SM, dvol_{Sg}) \xrightarrow{\Phi_{\pm}} \xrightarrow{\sim} L^2((0,1), L^2(M, \Sigma M, \sin^n(r)\, dvol_g))$$

s.t.

$$\Phi_- \circ D^+ \circ \Phi_-^{-1} = \partial_r + \frac{n}{2}\cot(r) + \frac{1}{\sin(r)} D_M$$

compose $\Phi_{\pm}$ with mult. with $\sin^{n/2}(r)$

$$(u,v)_{L^2} = \int_0^1 \int_M \langle u, v \rangle \sin^n(r)\, dvol_g\, dr$$

$$\rightsquigarrow L^2(U, \Sigma^{\pm} SM, dvol_{Sg})$$

$$\xrightarrow{\tilde{\Phi}_{\pm}} \xrightarrow{\sim} L^2((0,1), \underbrace{L^2(M, \Sigma M; dvol_g)}_{=: L})$$

and

$$\tilde{\Phi}_- \circ D^+ \circ \tilde{\Phi}_+^{-1} = \partial_r + \frac{1}{\sin(r)} D_M$$

(D_M: Dirac op. on M)

$$= \partial_r + \frac{1}{r} D_M + \underbrace{\left(\frac{1}{\sin(r)} - \frac{1}{r}\right) D_M}_{\text{pert. term}}$$

$\rightsquigarrow$ construct parametrix for $\mathcal{D}^+$

$$P : L^2(\Sigma^+ SM) \longrightarrow \mathrm{dom}(\overline{\mathcal{D}^+}) \quad \text{bounded s.t.}$$

$$P\,\overline{\mathcal{D}^+} = id + L \qquad \overline{\mathcal{D}^+}\,P = id + R$$

L, R: compact operators

existence of parametrix for $\overline{\mathcal{D}^+}$

$\Leftrightarrow$ $\overline{\mathcal{D}^+}$ Fredholm

deform. argument

$\Rightarrow$ $\exists$ harm. spinor

+ index formula of Chou '85

Furthermore, integrated SL-formula still holds for mfds. with cone-like singularities